MIL-P-20512D(EA)
12 May 1987
SUPERSEDING
MIL-P-20512C(EA)
14 February 1978

MILITARY SPECIFICATION

PROJECTILE, 155MM, SMOKE, HC

BE, M116A1, LOADING, ASSEMBLING AND PACKING

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the loading, assembling, and packing of Projectile, 155MM, Smoke, HC, BE, M116A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification For.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-1167 - Ammunition Data Card.
MIL-STD-1168 - Ammunition Lot Numbering.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

7549275 - Pallet for 155 MM Projectile.

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

8885162 - Projectile, 155 MM, Smoke, HC, BE, M116A1 Without Fuze.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Projectile. The projectiles shall comply with all requirements specified on Drawing 8885162, and applicable specifications.

3.2.1 Expelling charge assembly. There shall be no evidence of loose black powder in the fuze threads or fuze well area of the projectile after load,

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assemble and pack or evidence of a broken or cracked expelling charge assembly prior to insertion of the lifting plug. Evidence of either condition is considered a critical defect (see 4.2.3.2).

3.3 Proving ground. The projectiles shall comply with the following requirements when tested as specified in 4.3.1.

3.3.1 Functioning. The projectile shall eject its four canisters during flight. Each canister, after ejection, shall ignite and emit white smoke without explosion or disintegration for no less than 57 seconds to 170 seconds for the M1 canister and 27 seconds to 140 seconds for the M2 canister (see 4.3.1.1).

3.3.2 Premature functioning and security of parts. There shall be no premature functioning or indication of metal parts failure within the gun bore or in flight before functioning of the fuze.

3.3.3 Safety. There shall be no premature burst or indication of metal parts separation in the gun bore or in flight before functioning of the fuze when fired with the M4A2 propelling charge (zone 7) conditioning at 145° - 5°F, from a howitzer equipped with an M126 or M1A2 cannon. Observation shall be made for compliance with the specified requirements of section 4 (see 4.3.1.3 and 4.5.1.3).

3.4 Workmanship. The assembled projectile shall be free from contamination (oil, rust, grease, foreign material) and damage, including but not limited to; dents, cracks, or deformation which could affect its intended use. Care should be exercised at all times to prevent damage (nicks and burrs) to the projectile rotating band.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.1.2 Additional responsibility of contractor. Drawings for contractor designed inspection equipment (other than standard measuring equipment) shall be forwarded to the design agency for approval prior to production (see 6.1(b)). Ammunition data cards prepared in accordance with MIL-STD-1167 shall be supplied for each lot of projectiles.

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of section 3 and section 5, for which specific inspection has not been provided, have been satisfied.

4.2 Quality conformance inspection.

4.2.1 Lotting. Each projectile lot shall be identified and controlled in accordance with MIL-STD-1168. Each lot of projectiles shall contain:

(a) Bodies of one interfix number from one manufacturer.
(b) Parts of one interfix number from one manufacturer.
(c) Expelling charges of one interfix number from one manufacturer.
(d) Canisters of one interfix number from one manufacturer.
(e) Canisters containing reclaimed smoke mix shall not be mixed with canisters containing regular smoke mix.
(f) Loaded projectiles of one weight zone.

4.2.2 Sampling. Sampling of projectile lots shall be in accordance with MIL-STD-105, level II. The AQLs shall be as specified in 4.2.3.2. Continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the procuring activity.

PRECAUTION - This specification covers sampling and testing of chemical toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.2.3 Inspection procedure.

4.2.3.1 Examination and nondestructive tests. The sample projectiles shall be examined in accordance with the classification of defects (4.2.3.2) and MIL-STD-105.

4.2.3.2 Classification of defects. See pages 5 through 7.

4.3 Testing.

4.3.1 Proving ground. The following test shall be performed as specified in 4.5.1.

4.3.1.1 Functioning (see 3.3.1). Major defect - Beginning with the first lot produced and continuing until three consecutive lots have complied with the

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
4.2.3.2	Projectile, 155MM, Smoke HC, BE, M116A1, Assembling and Packing				8885162
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Projectile zone weight or marking incorrect by more than two weight zones		100%	3.2	Visual/scale
2	Black powder in fuze well area or threads		100%	3.2	Visual
Major					
1	Diameter of bourrelet, max (if retouched)		100%	3.2	Gage
2	Projectile zone weight or marking incorrect by two weight zones		100%	3.2	Visual/scale
Major					
101	Foreign material in cavity (prior to insertion of expelling charge assembly)		0.40%	3.2	Visual
102	Expelling charge missing		0.40%	3.2	Visual
103	Expelling charge support missing		0.40%	3.2	Visual
104	Packing tape not removed		0.40%	3.2	Visual
105	Component missing or incorrectly assembled (prior to assembly of base plug)		0.40%	3.2	Visual
106	Sealing compound missing from projectile threads (prior to assembly of base plug)		0.40%	3.2	Visual
107	Base plug assembled incorrectly		0.40%	3.2	Gage, torque wrench
108	Overall length, (max) incorrect		0.40%	3.2	Gage, torque wrench

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 3		DRAWING NUMBER 8885162
4.2.3.2	Projectile, 155 MM, Smoke, HC, BE, M116A1, Assembling and Packing				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
109	Pin missing or incorrectly assembled (above flush with outer surface)		0.40%	3.2	Visual & gage
110	Loose canister pack		0.40%	3.2	Gage
111	Lifting plug improperly assembled		0.40%	3.2	Torque wrench
112	Lifting plug gasket missing		0.65%	3.2	Visual
113	Projectile zone weight or marking incorrect by one weight zone		0.65%	3.2	Visual/scale
Minor					
201	Grommet missing damaged or incorrect		0.65%	3.2	Visual
202	Rotating band damaged		0.65%	3.2	Visual
203	Paint damaged (total damaged area in excess of 1/4 square inch)		0.65%	3.2	Visual
204	Marking missing or incorrect except for zone marking		0.65%	3.2	Visual
205	Color code incorrect		0.65%	3.2	Visual
206	Workmanship		0.65%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER 7549275
4.2.3.2	Pallet for 155 MM Projectile				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pallet damaged		0.65%	5.1	Visual
102	Federal Stock Number or DOD Identification Code missing or incorrect		0.65%	5.2	Visual
103	Projectile different zone weight on same pallet		0.65%	5.1	Visual
104	Strapping missing or loose		0.65%	5.1	Visual
105	Number of projectile assemblies incorrect		0.65%	5.1	Visual
106	Marking missing or incorrect		0.65%	5.2	Visual

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applicable requirements, 27 projectiles shall be randomly selected from each lot for the test. Twenty-five of the projectiles shall be fired for determination of satisfactory functioning (ejection, ignition, color and burning time of the 100 canister assemblies). If a total of nine or more canister assemblies fail to eject, ignite or burn for the required time, the lot shall be rejected. In the event that one or two projectiles fail to show evidence of fuze functioning, additional projectile(s) shall be fired in order that a total of 100 canister assemblies are observed for ejection, ignition, color, and burning time.

4.3.1.2 Reduced sampling for functioning. After three consecutive lots have met the criteria of 4.3.1.1, eleven projectiles shall be randomly selected from each lot for this test. Ten of the projectiles shall be fired for determination of satisfactory functioning (ejection, ignition, color, and burning time of the 40 canister assemblies). If a total of five or more canisters fail to ignite or burn for the required time, the lot shall be rejected. In the event that any of the projectiles fail to show evidence of fuze functioning, additional projectile(s) shall be fired in order that a total of 40 canister assemblies are observed for ejection, ignition, color and burning time.

4.3.1.3 Safety (see 3.3.3). Critical defect - Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements, five projectiles shall be randomly selected from each lot for this test. If any projectiles fail to comply with the requirements, the lot shall be rejected.

4.3.1.4 Reduced sampling for safety. After three consecutive lots have met the criteria of 4.3.1.3, three sample projectiles shall be randomly selected from each lot for the test. If any projectile fails to comply with the requirements, the lot shall be rejected.

4.3.1.5 Premature functioning and security of parts (see 3.3.2). Critical defects - If any of the sample projectiles used in any of the tests fails to comply with the applicable requirements, the lot shall be rejected.

4.3.1.6 Refiring. If for any reason the proof director considers that the test conditions have detrimentally affected the test results, he shall declare the results invalid and perform a new test on additional rounds as required.

4.4 Government use of contractor's inspection equipment. The contractor shall make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.5 Test methods and procedures.

4.5.1 Proving ground tests. These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure.

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4.5.1.1 Functioning. The sample projectiles shall be fired with a zone 7 M4A2 propelling charge in a howitzer equipped with an M126 or M1A2 cannon.

4.5.1.2 Safety. The sample projectile shall be fired with a zone 7 propelling charge (M4A2 conditioned at 145° - 5°F from a howitzer equipped with an M126 or M1A2 cannon. Observation shall be made for compliance with specified requirements (see 4.3.1.3 and 4.5.1.3).

4.5.1.3 Malfunctions.

(a) Misfires. The projectile fails to fire from the weapon.

(b) Premature functioning and metal parts failure. There shall be no premature burst or indication of metal parts failure in the gun bore or in the first 200 yards (182.88 meters) of horizontal range.

(c) Fuze dud. Absence of visible evidence of fuze functioning as evidenced by a flash or presence of a smoke cloud.

(d) Projectile dud. Canister payload failed to eject.

(e) Lost. Observer(s) fail to see canister ejection or canister burning.

(f) Canister dud. The canister is ejected but no smoking or burning occurs.

(g) Exploder. The canister is ejected and then explodes. This may occur with or without some normal smoke emission. It generally results from smoke mix anomaly and causes a black cloud at the point of incident. The metal parts commonly have the canister lid or bottom blown out, or both lid and bottom.

(h) Disintegrator. The canister is ejected and then splits open (usually into approximately equal segments) dispersing the smoke mix into the air. The cloud thus produced is the same color as the fill. The metal parts usually have the canister lid on the flash tube.

(i) Delayed ignition. The smoke emission from the canister did not begin within 12 seconds.

(j) Incorrect burn time. The canister smoke emission time is less than 57 or more than 170 seconds for the M1 canister and less than 27 or more than 140 seconds for the M2 canister.

(k) Poor smoke. HC smoke that does not obscure.

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(1) Flamer. Canister flaming greater than 10 percent of smoke emission time. Canisters that fall in tall, dry grass, will be observed very carefully to insure that the grass does not burn first and then flame the canister. HC canisters exhibit flame at each end during complete smoke emission time, but this is not considered to constitute flaming.

5. PACKAGING

5.1 Packing.

5.1.1 Level A. The projectiles shall be packed in accordance with Drawing 7549275. Projectiles of only one zone weight shall be packed on any one pallet.

5.2 Marking. Marking shall be in accordance with Drawing 7549275.

6. NOTES

6.1 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Design agency where drawings for contractor designed inspection equipment are to be forwarded.

6.2 Subject term (key word) listing.

Ammunition
Canister, smoke
Expelling charge
Projectile
Propelling charge

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.4 Component specification. The following specification covers the parts for the M116A1 projectile:

MIL-P-60340 - Projectile, 155MM, Smoke, HC, BE, M116A2, Parts For.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 1320-A962

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-20512D(EA)	PROJECTILE, 155MM, SMOKE, HC BE, M116A1, LOAD, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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